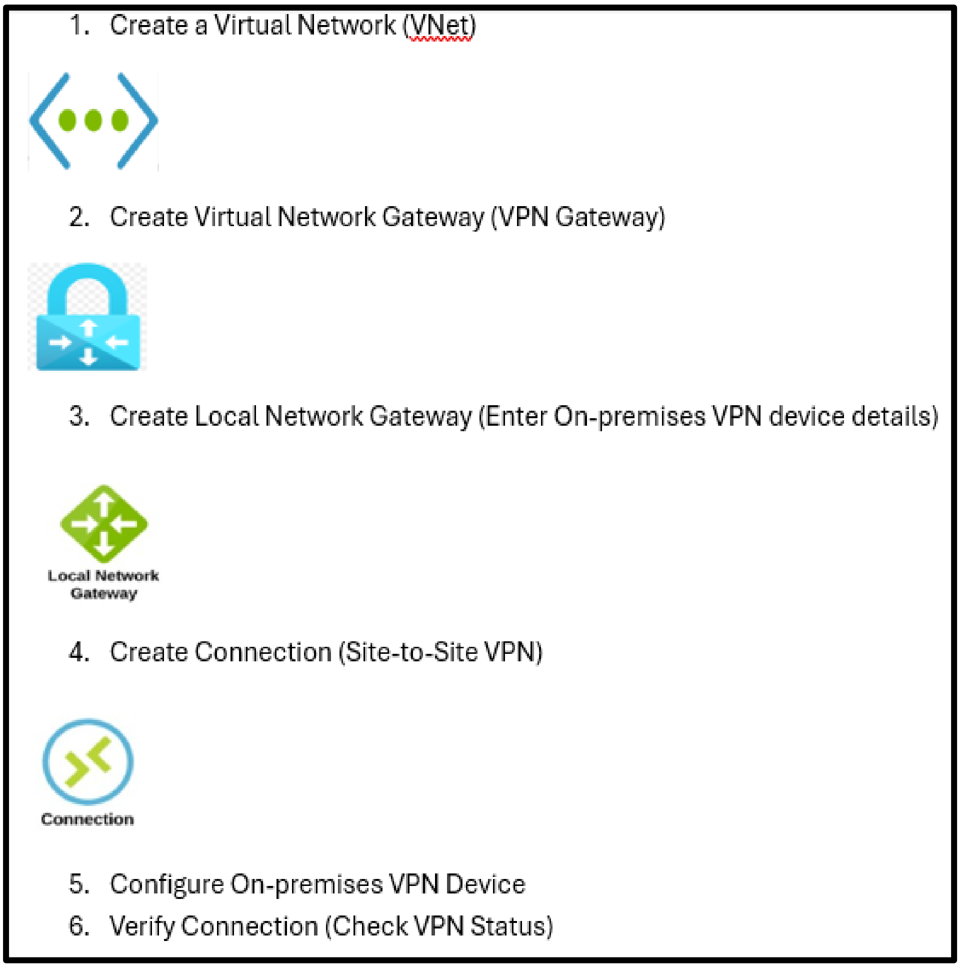
To create a **Site-to-Site VPN** in **Microsoft Azure**

Need to establish a secure connection between your on-premises network (or another cloud environment) and Azure virtual network (VNet). Here's a step-by-step guide to set this up:

Prerequisites

1. **Azure Subscription**: Need an Azure account.
2. **On-Premises VPN Device**: Need a compatible on-premises VPN device or gateway (like a Cisco ASA, Fortinet, or any supported VPN appliance).
3. **Public IP Address**: On-premises gateway should have a public IP address.
4. **Virtual Network (VNet)**: Need an Azure VNet to connect to.

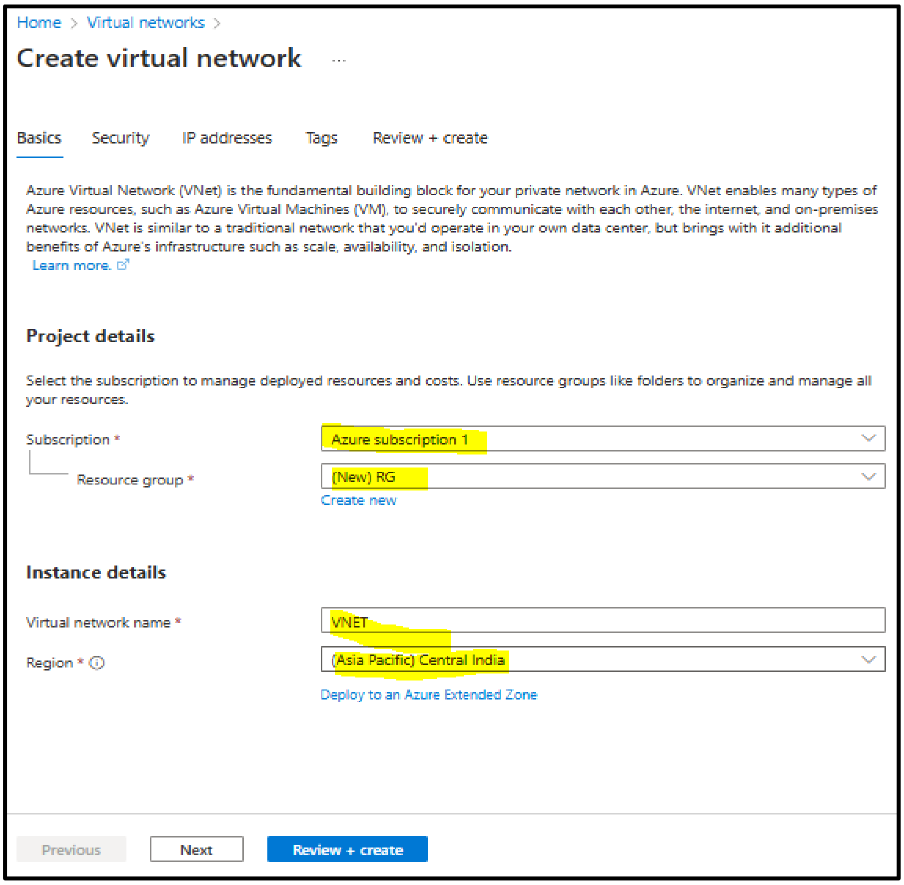
Flowchart Diagram Example



Steps to Create a Site-to-Site VPN in Azure

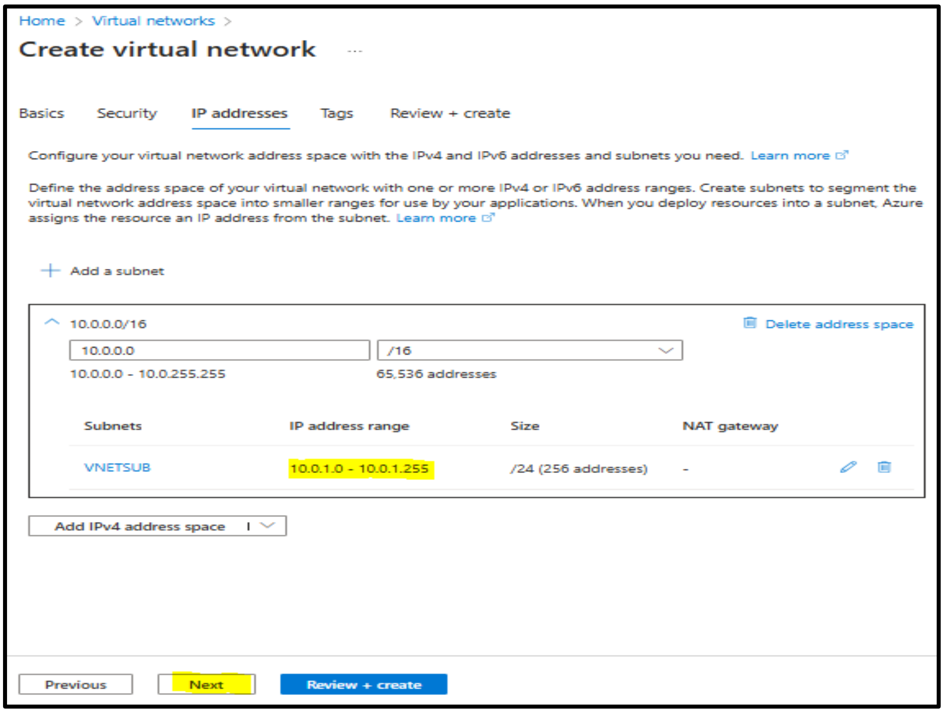
Step 1: Create a Virtual Network (VNet) in Azure

1. **Log into Azure Portal** (portal.azure.com).
2. In the left-hand menu, click **Create a resource**.
3. Search for **Virtual Network** and select it.
4. Click **Create**.
5. In the **Create Virtual Network** pane:
    - **Subscription**: Choose subscription.
    - **Resource group**: Choose an existing resource group or create a new one.
    - **Name**: Provide a name for the VNet (e.g., AzureVNet).
    - **Region**: Choose a region close to on-premises network.

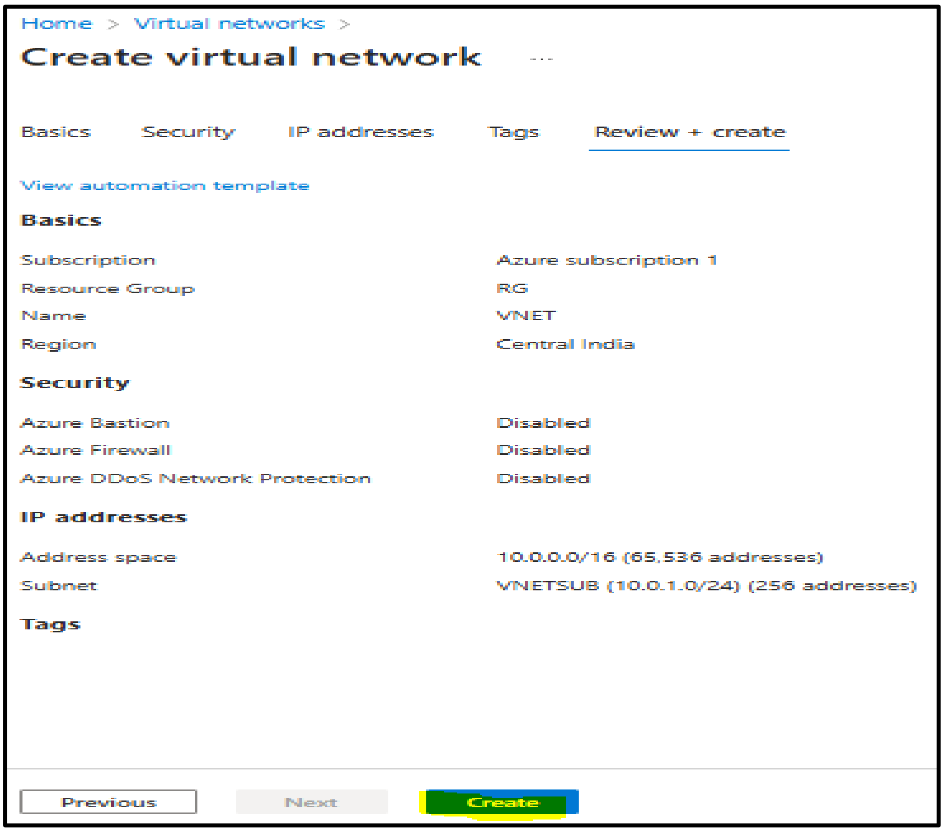


- **IP Addressing**: Define the address space (e.g., 10.0.0.0/16).

- **Subnets**: Create a subnet (e.g., 10.0.1.0/24).

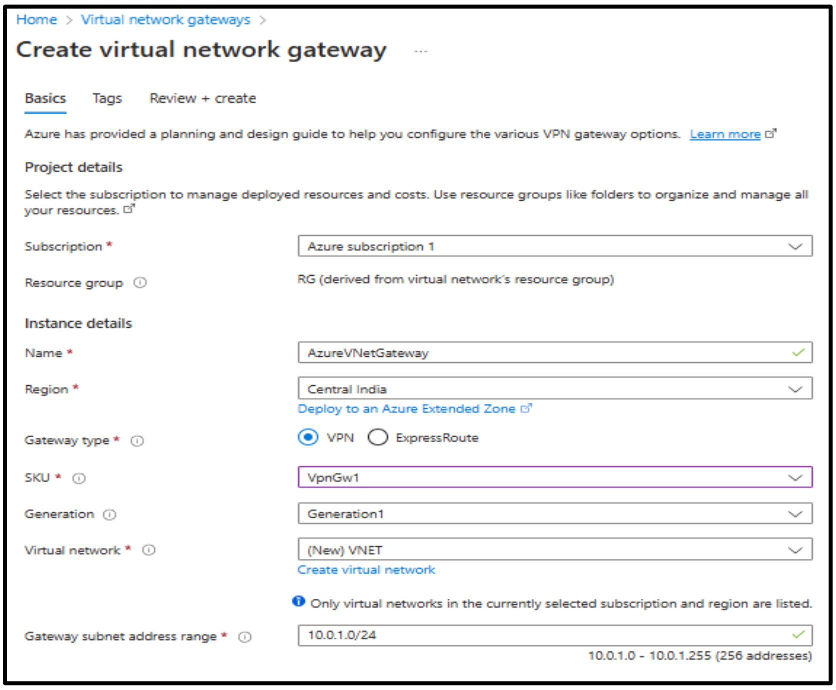


6. After reviewing the configuration, click **Create**.

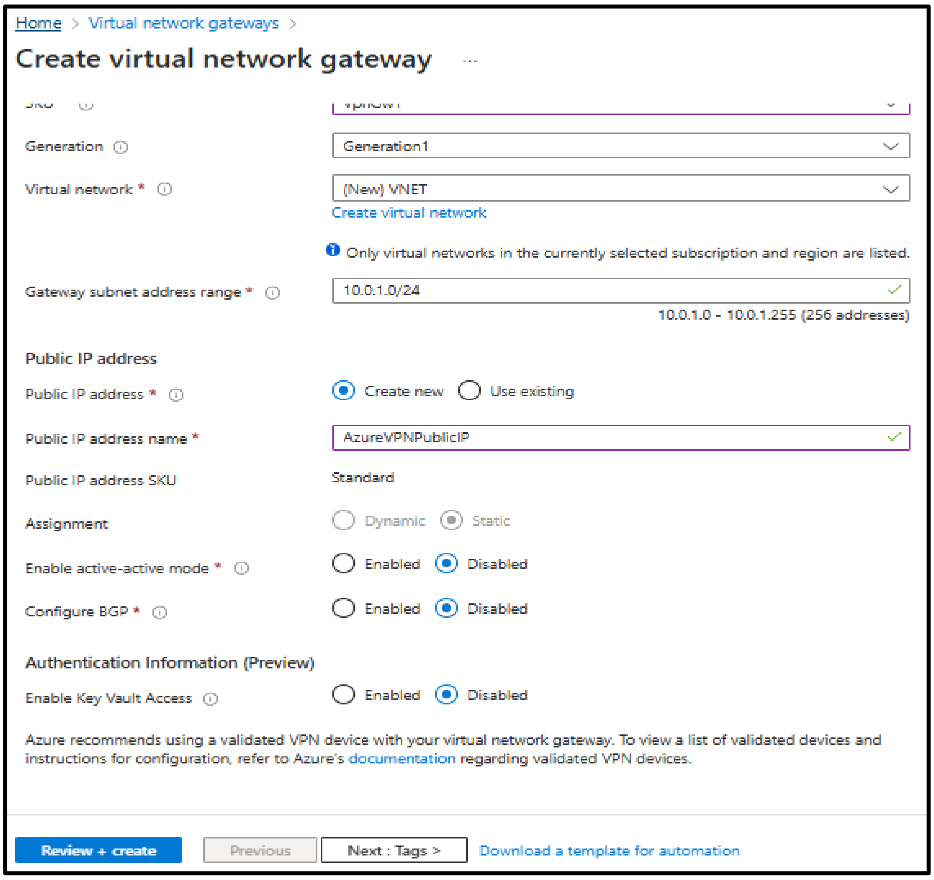


Step 2: Create a Virtual Network Gateway

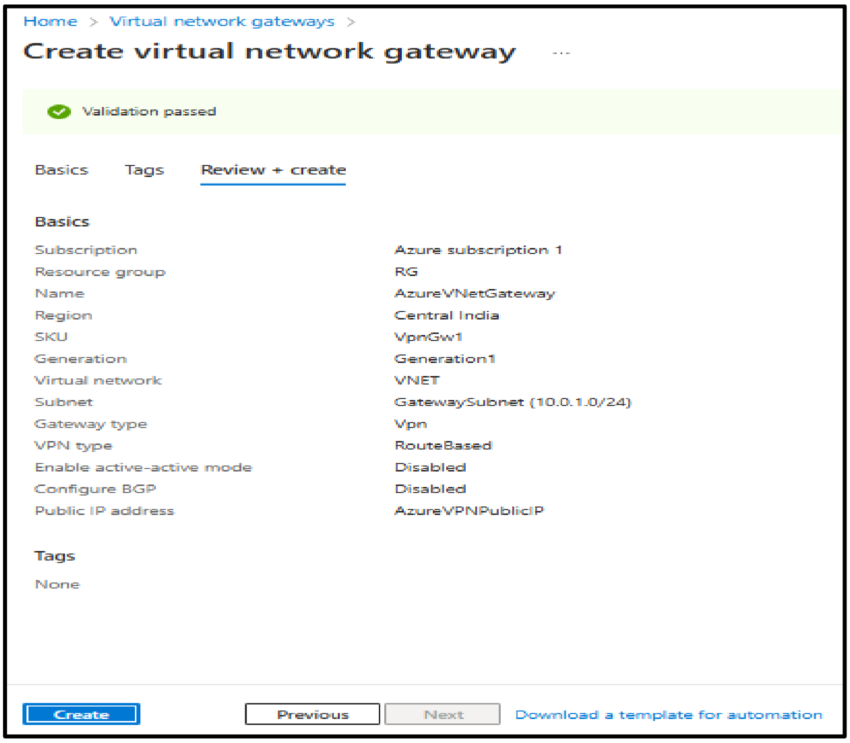
1. In the Azure portal, search for **Virtual network gateways**.
2. Click **Add** to create a new gateway.
3. In the **Create virtual network gateway** pane:
    - **Subscription**: Select the subscription where VNet is located.
    - **Resource Group**: Select the same resource group as VNet.
    - **Region**: Select the same region as VNet.
    - **Name**: Provide a name for the gateway (e.g., AzureVNetGateway).
    - **Gateway Type**: Select **VPN**.
    - **VPN Type**: Choose **Route-based** (recommended for most scenarios).
    - **SKU**: Select a SKU (e.g., **VpnGw1** or **VpnGw2** depending on needs).
    - **Virtual Network**: Select the VNet created earlier.



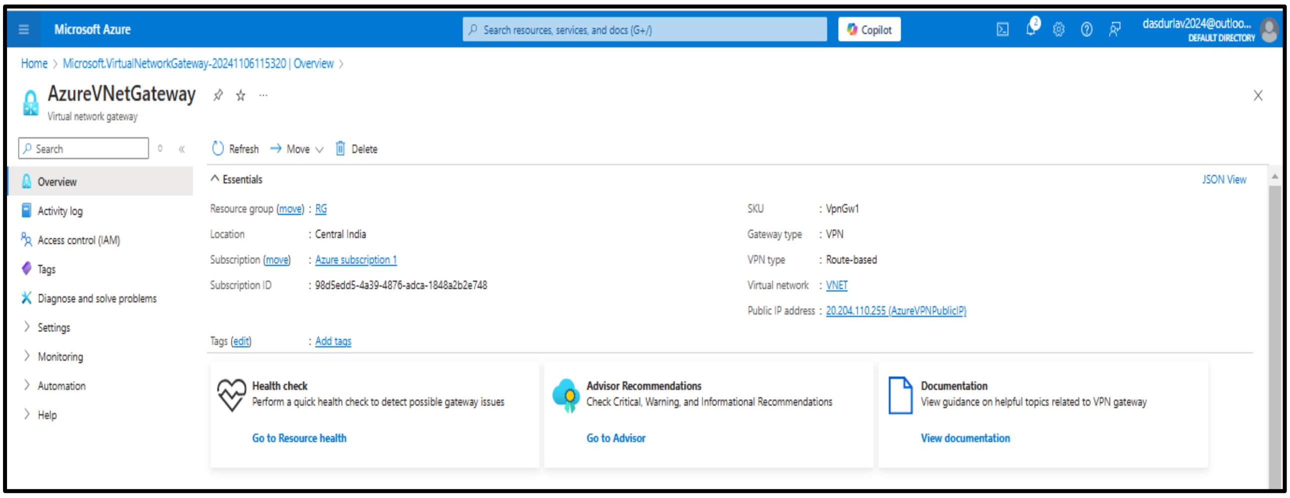
    - **Public IP Address**: Select **Create new** and give it a name (e.g., AzureVPNPublicIP).
    - **Configure BGP**: Choose **Disabled** unless you are using BGP for routing.



4. Click **Review + Create**, and after validation, click **Create**.

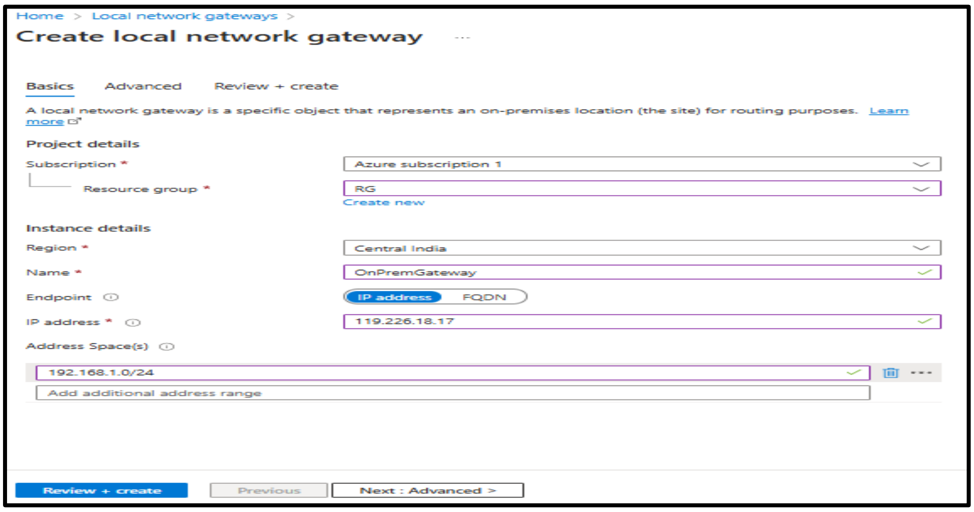


Note: The Virtual Network Gateway takes 30 minutes or more to deploy, depending on the SKU.

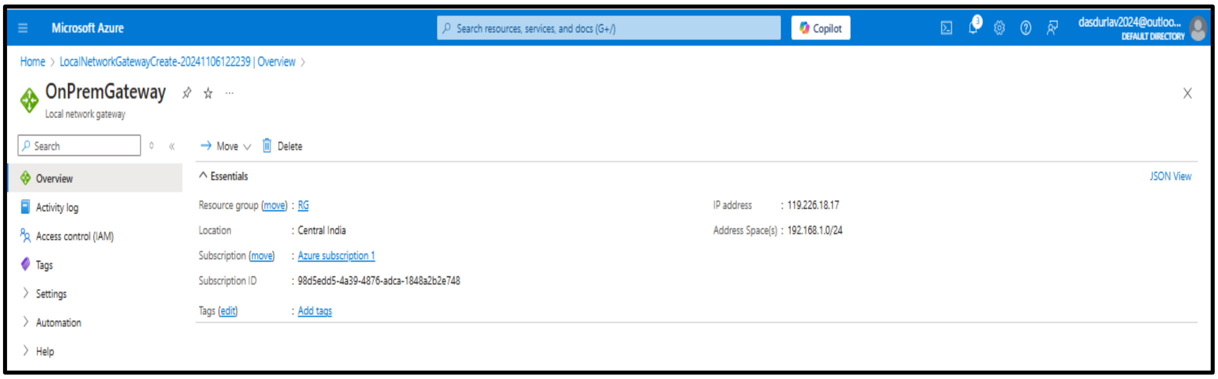


Step 3: Configure the Local Network Gateway (On-Premises VPN Device)

1. In the Azure portal, search for **Local network gateways**.
2. Click **Add** to create a new local network gateway.
3. In the **Create local network gateway** pane:
   - **Subscription**: Select the subscription.
   - **Resource Group**: Select the same resource group.
   - **Name**: Provide a name for the local network gateway (e.g., OnPremGateway).
   - **IP Address**: Enter the public IP address of on-premises VPN device.
   - **Address Space**: Provide the IP address range of on-premises network (e.g., 192.168.1.0/24).

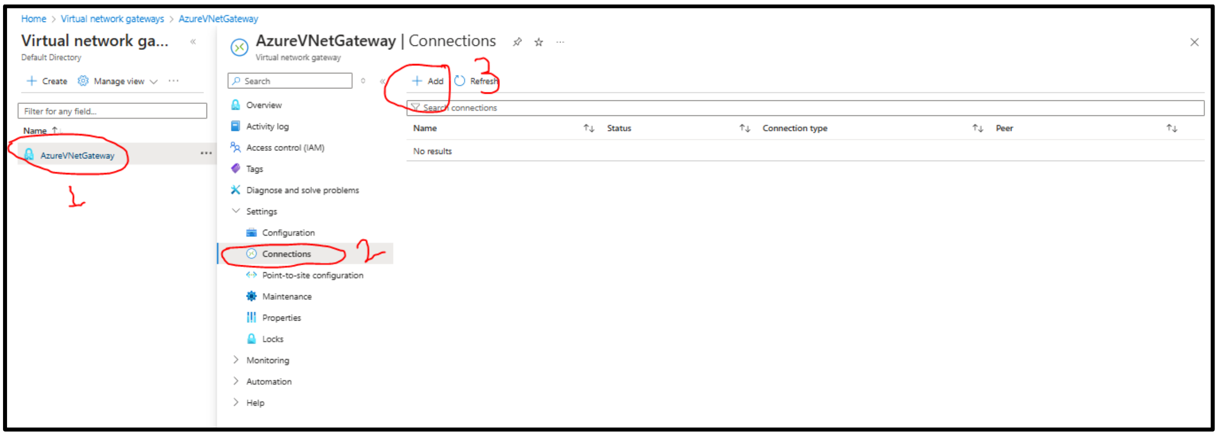


4. Click **Review + Create** and then **Create**.

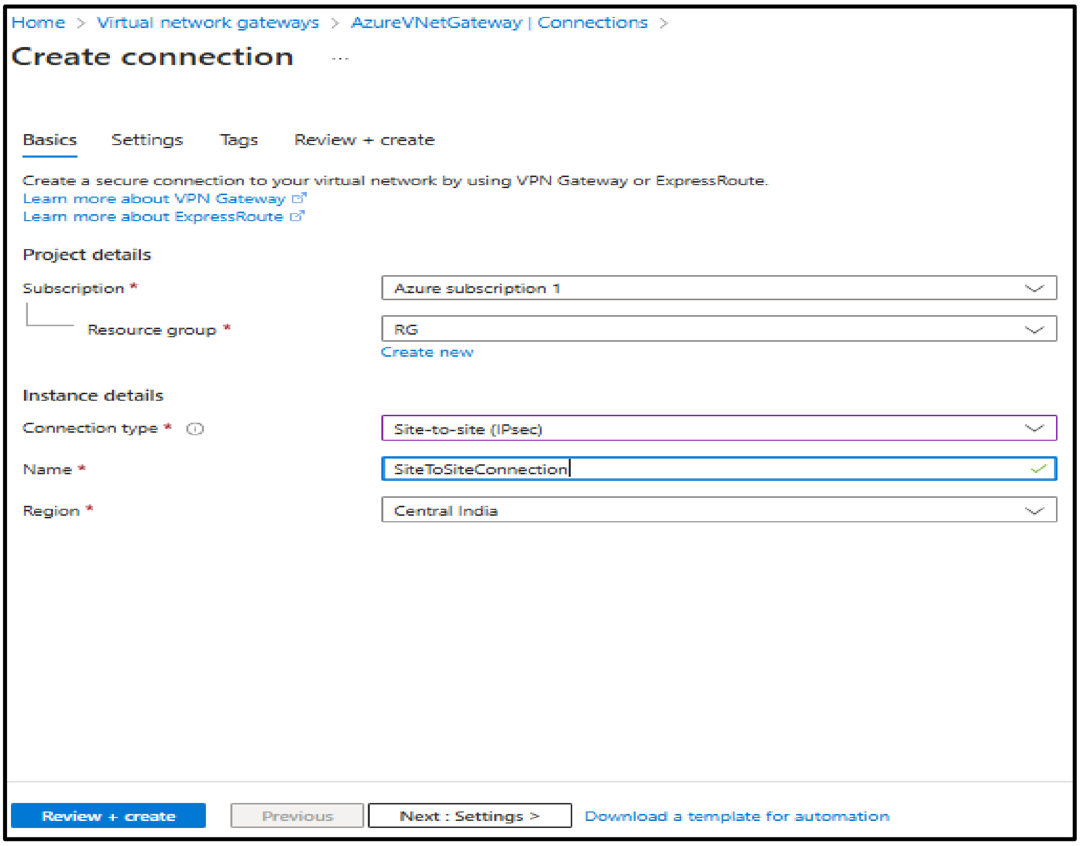


Step 4: Create the VPN Connection

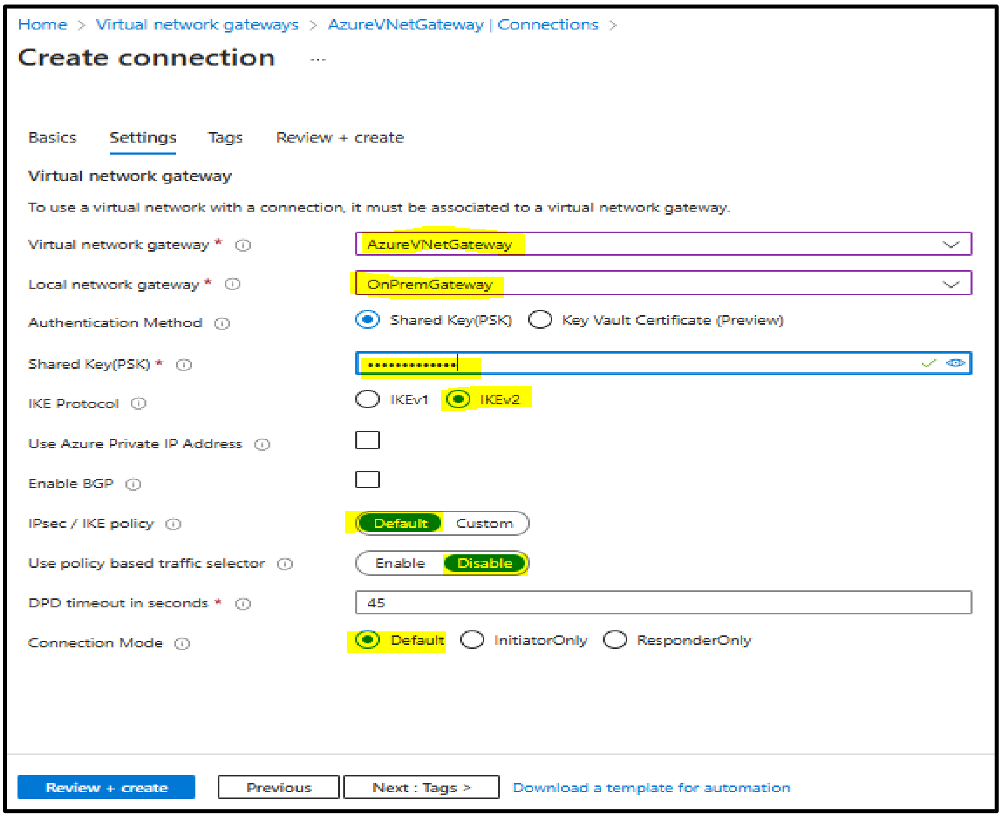
1. In the Azure portal, navigate to the **Virtual Network Gateway** you created earlier.
2. Under **Settings**, click **Connections**.
3. Click **Add** to create a new connection.



4. In the **Add connection** pane:
   - **Name**: Provide a name for the connection (e.g., SiteToSiteConnection).
   - **Connection Type**: Select **Site-to-Site (IPsec)**.
   - **Virtual Network Gateway**: This is pre-populated with your virtual network gateway.
   - **Region**: Select region



- **Local Network Gateway**: Select the local network gateway (you're on-premises device) you created earlier.
- **Shared Key**: Enter a shared secret (pre-shared key or PSK) that is also configured on your on-premises VPN device.
- **IKE Protocol**: Select IKE version (IKEv1/IKEv2)
- **IPsec/IKE policy**: Select policy mode (Default/Custom)
- **Policy based Traffic**: Enable/Disable the policy-based traffic selector.
- **Connection Mode**: Select connection mode (Default/Initiator only/ Responder only)


Home > Virtual network gateways > AzureVNetGateway | Connections >
Create connection ...
Basics
Settings
Tags
Review + create
Virtual network gateway
To use a virtual network with a connection, it must be associated to a virtual network gateway.
Virtual network gateway *
AzureVNetGateway
Local network gateway *
OnPremGateway
Authentication Method
Shared Key(PSK)
Key Vault Certificate (Preview)
Shared Key(PSK) *
IKE Protocol
IKEv1
IKEv2
Use Azure Private IP Address
Enable BGP
IPsec / IKE policy
Default
Custom
Use policy based traffic selector
Enable
Disable
DPD timeout in seconds *
45
Connection Mode
Default
InitiatorOnly
ResponderOnly
Review + create
Previous
Next : Tags >
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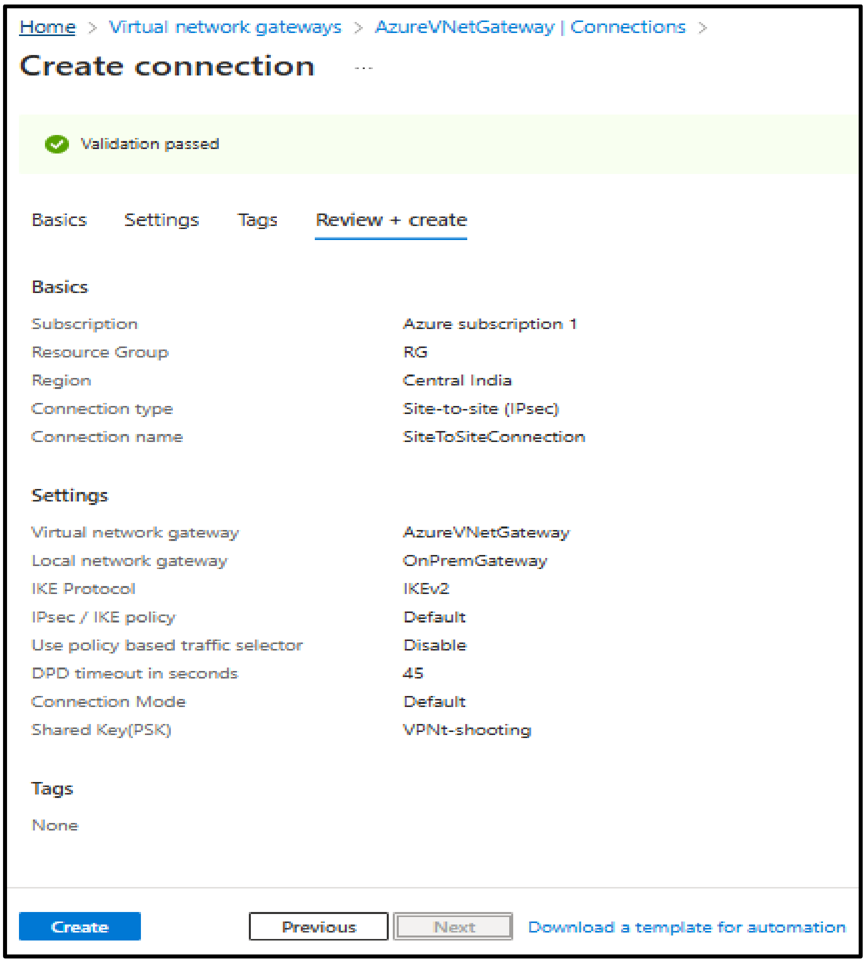
- Click **Next.**


Home > Virtual network gateways > AzureVNetGateway | Connections >
Create connection ...
Validation passed
Basics
Settings
Tags
Review + create
Basics
Subscription
Azure subscription 1
Resource Group
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Connection type
Site-to-site (IPsec)
Connection name
SiteToSiteConnection
Settings
Virtual network gateway
AzureVNetGateway
Local network gateway
OnPremGateway
IKE Protocol
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IPsec / IKE policy
Default
Use policy based traffic selector
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45
Connection Mode
Default
Shared Key(PSK)
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None
Create
Previous
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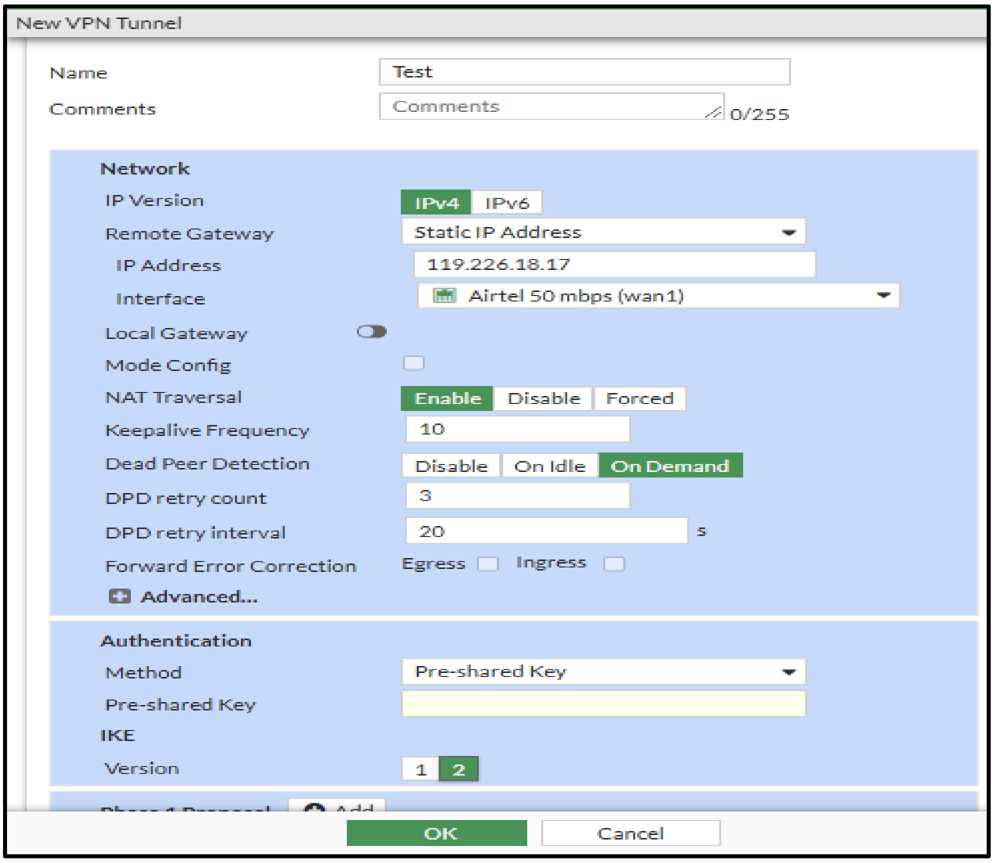
- Click **Review + Create** and then **Create**

Step 5: Configure Your On-Premises VPN Device

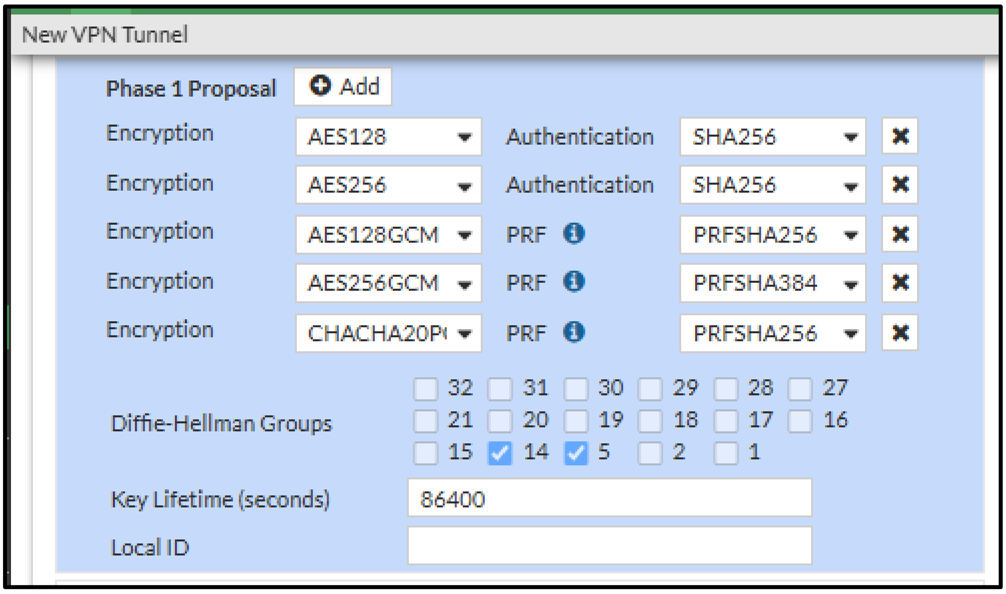
Configure on-premises VPN device to establish the IPsec tunnel with Azure. The specific configuration depends on the device you're using, but here's an overview of the typical settings:

- **VPN Type**: IPsec/IKEv2
- **Gateway IP**: Public IP of the Azure VPN gateway.
- **Shared Key**: The pre-shared key (PSK) you specified in the Azure VPN connection.
- **Local Network**: on-premises subnet (e.g., 192.168.1.0/24).
- **Remote Network**: The Azure VNet's subnet (e.g., 10.0.1.0/24).
- **IKE Version**: Version 2 (IKEv2).
- **IPsec Encryption**: Ensure the encryption settings match what is supported by Azure (e.g., AES256).

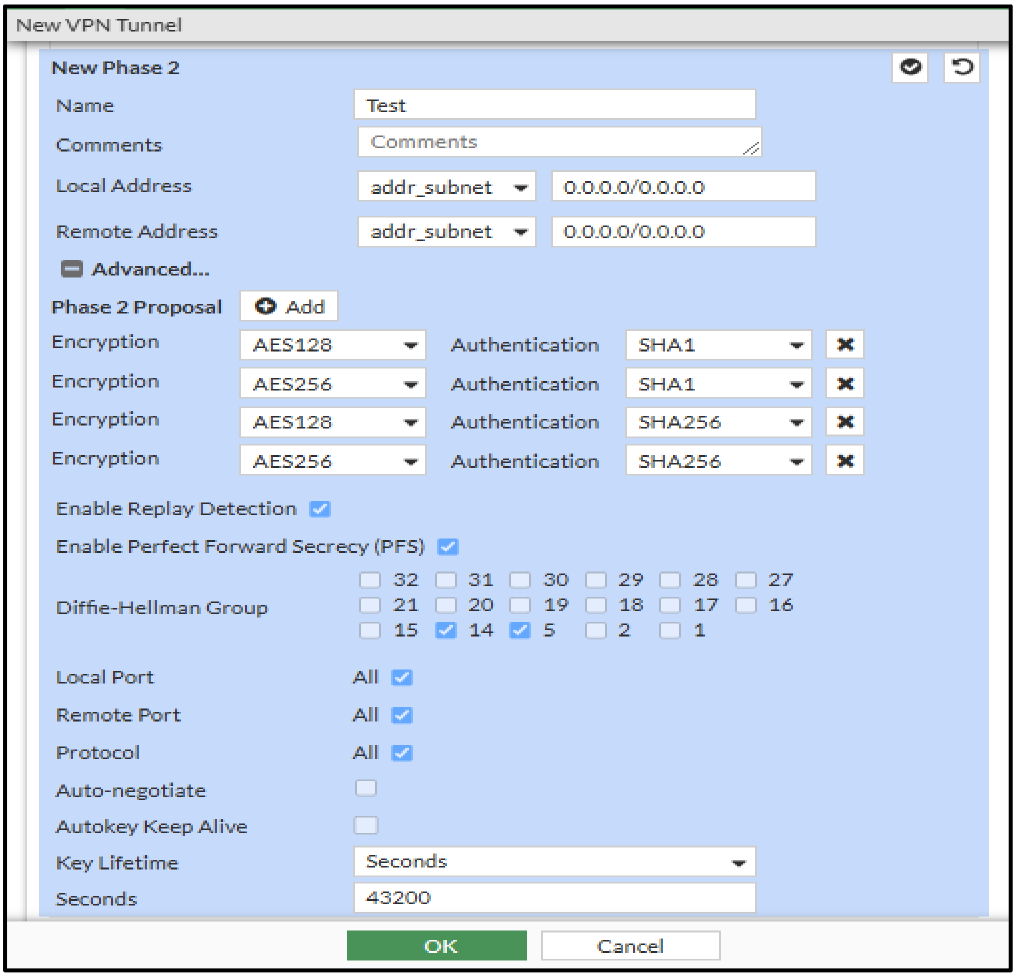
For Example, FortiGate Firewall showcase here



Phase 1 configuration



Phase 2 configuration



Note: Select Phase 1 and Phase 2 configuration as per Azure end. If not then connection not stablished.

Check your VPN device's documentation for detailed setup steps.

Step 6: Test the VPN Connection

Once everything is set up, check the connection status in the **Azure portal** under the **Connection** tab in the Virtual Network Gateway.

- **Status**: The connection should show as **Connected**.
- To verify connectivity, try to ping a resource in Azure (e.g., a virtual machine in the VNet) from your on-premises network.

Troubleshooting

- Ensure that the shared key (PSK) matches on both sides.
- Double-check the IP address ranges for both Azure and on-premises networks to avoid overlap.
- Check firewall rules to ensure that UDP ports 500 and 4500 are open for IKE/IPsec negotiation.
- If using BGP, make sure BGP settings are configured correctly on both sides.

Optional Steps: Configure Route Tables (if needed)

1. **On-Premises Routing**: Ensure that on-premises network has a route to reach the Azure VNet via the VPN tunnel.
2. **Azure Route Tables**: Need to configure route tables in Azure to properly route traffic from VNet to the on-premises network.